

Lembar Kerja 2

Kelompok: 01

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Preparation

Install

React, Chart.js, MQTT.js

Dashboard on React

```
App.js  x  {} package.json  main.cpp  PIO Home
ecg-dashboard > src > JS App.js > App

25 const MAX_POINTS = 100;
26
27 function App() {
28   const [status, setStatus] = useState("Disconnected");
29   const [heartRate, setHeartRate] = useState(0);
30   const [ecgBuffer, setEcgBuffer] = useState([]);
31   const clientRef = useRef(null);
32
33   useEffect(() => {
34     const client = mqtt.connect("wss://91c95e4b1fc04b6f9eb2e90b69199494.s1.eu.hivemq.cloud:8884/mqtt", {
35       username: "hivemq.webclient.1759562092546",
36       password: "7cluE#yd956o4RgF:VG"
37     });
38     clientRef.current = client;
39
40     client.on("connect", () => {
41       setStatus("Connected");
42       client.subscribe("ecg/data");
43     });
44
45     client.on("reconnect", () => setStatus("Reconnecting..."));
46     client.on("close", () => setStatus("Disconnected"));
47
48     client.on("message", (topic, message) => {
49       try {
50         const data = JSON.parse(message.toString());
51         setHeartRate(data.hr);
52         setEcgBuffer(prev => {
53           const next = [...prev, data.ecg];
54           return next.length > MAX_POINTS ? next.slice(-MAX_POINTS) : next;
55         });
56       } catch (e) {}
57     });
58
59     return () => client.end();
60   }, []);
61
62   const chartData = {
63     labels: Array(ecgBuffer.length).fill(""),
64     datasets: [
65       {
66         label: "ECG",
67         data: ecgBuffer,
68         borderColor: "#00ff00",
69         backgroundColor: "rgba(0,255,0,0.1)",
70         tension: 0.3,
71         pointRadius: 0,
```

Scenario

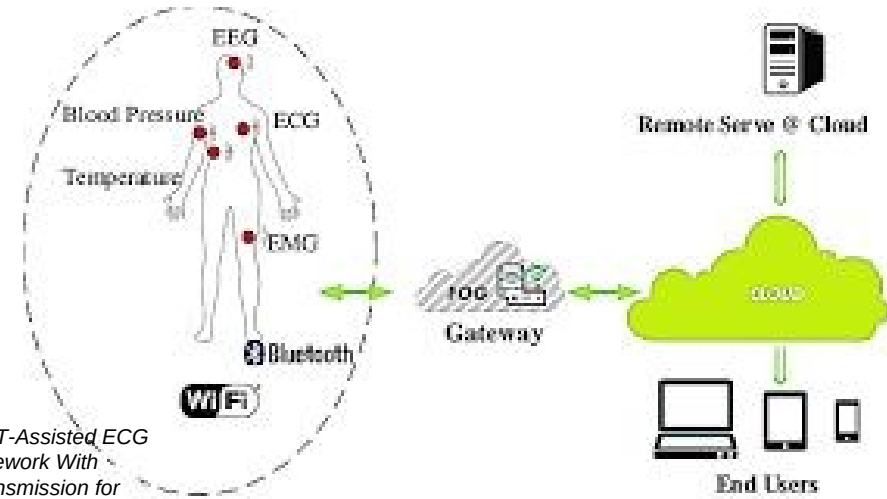
Kita akan mensimulasikan satu channel ECG sinyal didalam ESP32.

dalam 1 detik:

$1000 \text{ ms} / 8 \text{ ms} \approx 125 \text{ data}$

dipublish ke topic ecg/data setiap detik.

Kita publish datanya ke cloud



Xu, G. (2020). *IoT-Assisted ECG Monitoring Framework With Secure Data Transmission for Health Care Applications*. IEEE Access, 8, 74586-74594.

Program react



ESP32 (publisher)

MQTT Payload structure

```
StaticJsonDocument<200> doc;  
doc["ecg"] = ecgValue;           // Nilai ECG (float, hasil simulasi + noise)  
doc["hr"] = heartRate;           // Heart rate (int, acak antara 60-100 setiap siklus)  
doc["timestamp"] = currentMillis; // Waktu pengiriman (millis)  
  
char buffer[256];  
serializeJson(doc, buffer);
```

Questions:

Little-Endian

Byte paling tidak signifikan disimpan di alamat memori terkecil

Seperti membaca dari kanan ke kiri: byte paling tidak penting dulu

Contoh: Angka 0x12345678

Big-Endian

Byte paling signifikan disimpan di alamat memori terkecil

Seperti membaca dari kiri ke kanan: byte paling penting dulu

Contoh: Angka 0x12345678

Batch-mode adalah metode pengiriman data di mana beberapa data dikumpulkan dan

Questions:

Alasan Utama Menggunakan Integer vs Float

1. Efisiensi Pemrosesan (Processing Efficiency)

Integer operations jauh lebih cepat di microcontroller

Float operations membutuhkan FPU (Floating Point Unit) yang tidak dimiliki banyak microcontroller murah

Contoh perbandingan:

Integer multiplication: 1-2 clock cycles

Float multiplication: 10-20+ clock cycles

2. Konsistensi dan Presisi (Consistency & Precision)

Floating point memiliki masalah presisi (rounding errors)

Integer memberikan representasi yang eksak dan konsisten